

PRITISH SAHA

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Education

Indian Institute of Technology, Kharagpur

Kharagpur, West Bengal

B.Tech. in Manufacturing Science & Engineering + M.Tech. in Industrial & Systems Engineering

Expected completion: May'27

Courses: Safety Fundamentals of Generative AI | Operations Research | Probability & Statistics | Linear Algebra

Research & Engineering Experience

Research Fellow (MARS 4.0) | Cambridge AI Safety Hub | UK (Hybrid) | [\[Report\]](#)

Dec'25 – Present

With Prof. Fernando Rosas, studying Bayesian belief representations in transformers and extending experiments to RL

- Decoded beliefs from four-layer transformers in fully observed pilots: probe $R^2 = 0.985\text{--}0.997$ after 100k updates
- Built **exact Bayesian filters** to track hidden-state uncertainty in streams with complete or partial observations
- Implemented **exact POMDP solvers** and reward-switch experiments; 115 tests cover filtering and control

Data Science Intern | Axis Bank, Business Intelligence Unit | Mumbai | [\[Report\]](#)

May'26 – Jul'26

Built a PySpark pipeline to identify personal-loan fraud rings across 224.8M accounts and 1.31B transfer edges

- Pruned the working graph $\sim 20\times$ with time-respecting 3-hop BFS, retaining connections for $\sim 80\%$ of applicants
- Delivered a four-signal fraud review queue, with **5.49 $\times$ mean lift** in the **top 1%** across three monthly cohorts

Research Intern | RAAPID INC | Remote | [\[Research\]](#) [\[Patent\]](#) [\[LoR\]](#)

Mar'25 – Jul'26

With Prof. Amitava Das, developed efficient LLM fine-tuning methods and quality checks for clinical entity extraction

- Prototyped **Triton kernels** for covariance, matrix inversion and gradient preconditioning in LoRA fine-tuning
- Integrated **CUDA streams** and events to schedule covariance inversions and eigensolves, with PyTorch fallbacks
- Applied cleanup rules to 5,035 stored clinical predictions: **0.640** exact micro-F1 and **zero subword fragments**

Research Intern | Complex Networks Research Group, IIT Kharagpur | [\[Report\]](#) [\[LoR\]](#)

Jul'25 – May'26

With Prof. Pawan Goyal, studied latent alignment and reward shaping for mathematical reasoning in language models

- Added **latent alignment losses** to GRPO for Qwen2.5-Math-7B with LoRA: 16 rollouts/prompt on one H100
- Selected teacher-augmented pilot: **79.57%** MATH-500 mean accuracy (8 samples/problem; **+4.70 pp** vs GRPO)
- A separate audit found shaped credit aligned with GRPO (cosine **0.979**); no shaping signal on all-wrong groups

Publications & Preprints

Mixture of Chapters | Accepted, ICLR 2026 NFAM Workshop | [\[Code\]](#) | [\[Paper\]](#)

Mar'26

Tasmay Pankaj Tibrewal, **Pritish Saha**, Ankit Meda, Kunal Singh, Pradeep Moturi

- Co-developed sparse transformer memory; pretrained **150M–1B-parameter models** on up to **15B tokens**
- Limited ARC-C accuracy loss after instruction tuning to **2.68 pp** vs **6.69 pp** for a compute-matched baseline

GRIT: Geometry-aware PEFT | arXiv Preprint | [\[Code\]](#) | [\[Paper\]](#)

Jan'26

Pritish Saha, C. Rajbangshi, R. Goyal, M. Goyal, A. Deo, B. Roy, N. D. Singh, R. Goswami, A. Das

- Designed a LoRA method with **rank-space preconditioning**, Fisher-guided reprojection and adaptive ranks
- Scored **66.19%** on **GSM8K** vs LoRA's 63.38% on Llama-3.1-8B with **27.8% fewer** effective adapter parameters

Competitions

3rd Place | General Championship Data Analytics, IIT Kharagpur | [\[Report\]](#)

Mar'26

Captained Patel Hall's team in building a conversational analytics dashboard for database queries and KPI discovery

- Built **text-to-SQL** with retrieval and repair: **90% execution accuracy** on a 50-query BIRD mini-dev sample
- Automated KPI discovery with a LangGraph pipeline; generated **874K synthetic rows** across 58 client profiles

4th Place | ISRO GeoNLI, Inter IIT Tech Meet 14.0 | [\[Report\]](#)

Oct'25 – Dec'25

Co-designed a remote-sensing assistant for captioning, grounding and question answering across RGB, SAR and IR

- Designed a Qwen3-VL-8B curriculum with **LoRA SFT** and **DPO** for satellite captioning and question answering
- Paired **SAM3** with pyramidal tiling and VLM refinement for object counts, area estimates and spatial queries

National Finalist | American Express Campus Challenge, Decision Science | [\[Report\]](#)

Jul'25

Built a personalized offer-ranking system from customer histories, combining tree models with a transformer

- Reached **0.59 MAP@7** on final evaluation; trained a transformer to correct errors from LightGBM and XGBoost
- Generated **3K+ history features** with point-in-time joins and customer-grouped validation to limit leakage

Technical Skills

Languages & Systems: Python, C++, SQL, CUDA, Triton, PySpark, Linux, Docker, SLURM, Git

ML & Post-training: PyTorch, JAX, Transformers, LoRA/QLoRA, TRL, vLLM, FSDP/DTensor, FlashAttention-2

Extra-Curricular Activities

Karate black belt and tournament winner | Patel Hall interhall football and water polo | NSS village volunteer